

# COLLAPSE-ON-COMMAND OF AN ACTIVELY SUPPORTED ROTATING WORMHOLE: HORIZON FORMATION AND GRAVITATIONAL-WAVE EMISSION WITHOUT AN IMPOSED QUADRUPOLE KICK

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We evolve a rotating, actively supported wormhole in three-dimensional numerical relativity. Constraint-solved initial data contain an exotic complex scalar Q-torus with winding  $m = 1$ ; collapse is triggered by ramping its support to zero over  $t \in [8, 10]$ , without an imposed perturbation. The lapse reaches  $\min \alpha \simeq 10^{-3}$ , the conformal factor pinches, and a coordinate-sphere trapped-surface proxy grows to  $r_{\text{AH}} \simeq 9.2$ . The proxy then shrinks while the lapse recovers, indicating a phantom-matter rebound rather than a quiescent black hole. A coherent  $\ell = 2$  curvature wave propagates at approximately the speed of light and is overwhelmingly  $m = 0$ , linking the radiation primarily to the collapsing toroidal density rather than to a nonaxisymmetric spin mode. The collapse, rebound, and burst reproduce for base spacings  $h \simeq 0.667, 0.5, 0.4$ . Richardson behavior is quantity-dependent: selected geometric and momentum-constraint measures converge, whereas the Hamiltonian maximum and waveform peak do not admit controlled extrapolation. A completed  $L = 128$  run gives an independent intermediate-zone waveform check. The result is an engineered dynamical experiment, not a stationary solution of the unforced Einstein–Klein–Gordon system.

## I. INTRODUCTION

Traversable wormholes require null-energy-condition (NEC) violation [2]. Static Ellis–Bronnikov throats are dynamically unstable: compressive perturbations collapse them, while rarefactive perturbations drive expansion [4–7]. We ask whether support can be withdrawn from a rotating exotic throat in a controlled way and whether the collapse radiates without an imposed quadrupolar kick.

The rotation is carried by a complex scalar Q-torus rather than prescribed through an analytic rotating metric. An elliptic solve supplies constraint-consistent geometry and extrinsic curvature; an external support coupling holds the composite configuration temporarily and provides the trigger. This construction isolates the response to support removal but explicitly introduces a nonconservative source while support is active.

## II. METHOD

### A. Evolution system and initial data

We use the 3 + 1 line element

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt) \quad (1)$$

and evolve the CCZ4 system with `GRTeclyn` [9, 10]. The conformal decomposition is

$$\gamma_{ij} = \chi^{-1} \tilde{\gamma}_{ij}, \quad \det \tilde{\gamma}_{ij} = 1, \quad (2)$$

with 1 + log slicing and a Gamma-driver shift. A collapsing lapse diagnoses strong gravitational collapse;  $\chi \rightarrow 0$  marks large metric stretching.

The seed slice is a puncture representation of an Ellis–Bronnikov-class throat, not the analytic Teo metric [8]. `GRTresna` solves the Hamiltonian and momentum constraints for the exact throat-plus-matter configuration and exports a uniform `.gridinit` at the evolution base spacing. The evolution is never initialized at finer spacing than this solve.

The exotic complex scalar has potential

$$V(|\Phi|^2) = \frac{\mu^2}{2} |\Phi|^2 - \frac{\lambda}{4} |\Phi|^4 + \frac{\mu_6}{6} |\Phi|^6 \quad (3)$$

with  $\mu^2 = 0.5$ ,  $\lambda = 170$ , and  $\mu_6 = 1.445 \times 10^4$ . Its stationary flat-space profile is

$$\Phi = f(\rho, z) e^{i(m\varphi - \omega t)}, \quad m = 1, \quad \omega = 0.2507. \quad (4)$$

The solved torus is shown in Fig. 1. Curvature and self-gravity detune this profile after embedding, so the complete gravitating object is actively supported rather than a passive equilibrium.

### B. Trigger and numerical setup

The support strength is ramped from one to zero as

$$S(t) = \frac{1}{2} \left[ 1 + \cos \left( \pi \frac{t-8}{2} \right) \right], \quad 8 \leq t \leq 10, \quad (5)$$

with  $S = 0$  thereafter. No monopolar or quadrupolar field perturbation is added.

The headline domain has side  $L = 64$ , base spacing  $h = 0.5$ , and three AMR levels. A sponge occupies  $r = 28$ –30. The resolution ladder uses matched `GRTresna`

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Initial exotic Q-torus:  $\Phi = f(\rho, z)e^{i(m\varphi - \omega t)}$ ,  $m = 1$ ,  $\omega = 0.2507$

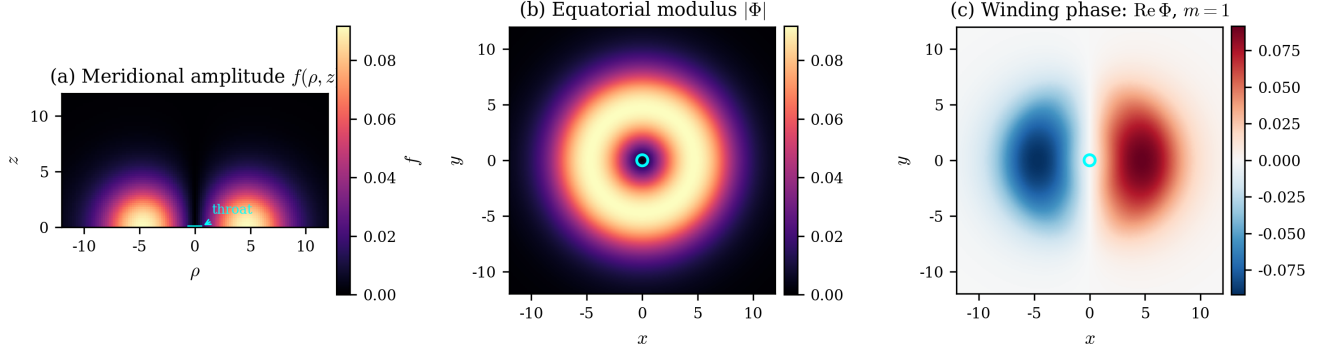


FIG. 1. Initial exotic matter. (a) The solved meridional eigenfunction  $f(\rho, z)$ . (b) Its axisymmetric toroidal modulus on the equatorial plane. (c) The real field at  $t = 0$ , showing the  $m = 1$  phase winding. The cyan circle/segment marks the throat scale. Thus the density is axisymmetric although the complex phase carries angular momentum.

data at  $h = 0.667, 0.5, 0.4$ , with otherwise identical evolution parameters. The large-box check uses  $L = 128$ ,  $h = 0.5$ , two AMR levels, a sponge at  $r = 52\text{--}62$ , and runs to  $t = 80$ .

Because `GRTEclyn` has no production apparent-horizon finder, we use the coordinate-sphere outgoing-expansion proxy

$$\theta_+ = \frac{2\sqrt{\chi}}{\bar{r}} - \frac{\partial_{\bar{r}}\chi}{\sqrt{\chi}} + \tilde{A}_{rr} - \frac{2}{3}K. \quad (6)$$

The largest sampled radius with  $\theta_+ \leq 0$  defines  $r_{\text{AH}}$ . This is a trapped-surface indicator, not a horizon mass or spin measurement.

Surface integrals give  $M_{\text{ADM}} = 0.209 \pm 0.001$  and  $J_{\text{ADM},z} \simeq -1.93$  in code units. We do not use these values to make precision detector forecasts because extraction systematics dominate.

### C. Wave extraction

We project  $\Psi_4$  onto  $_{-2}Y_{\ell m}$  on spheres at  $R_{\text{ext}} = 12, 18, 24$ . Physics results use the post-processed Python extraction from plotfiles; its files already store  $r\Psi_4^{2,0}$ . The in-code C++ Weyl extraction is retained only for debugging because its  $m = 0$  imaginary component fails the expected null test. In contrast, the Python signal has  $\max|\text{Im}\Psi_4^{2,0}|/\max|\text{Re}\Psi_4^{2,0}| \lesssim 1.2 \times 10^{-3}$ , and more than 99.998% of integrated  $\ell = 2$  power lies in  $m = 0$ .

The headline radii are in the near zone. The  $L = 128$  run extracts at  $R_{\text{ext}} = 36, 48$ , both inside the sponge onset, and is therefore an intermediate-zone check rather than a wave-zone extrapolation.

## III. RESULTS

### A. Collapse and rebound

Once support vanishes, the lapse and conformal factor collapse while the trapped-surface proxy grows (Fig. 2). In the headline run,

$$\min \alpha = 9.8 \times 10^{-4}, \quad \min \chi = 0.0185, \quad \max r_{\text{AH}} = 9.18.$$

The robust proxy turns on after the support cut and peaks near  $t = 15.1$ . Thereafter  $r_{\text{AH}}$  shrinks to 1.92 by  $t = 40$ , while  $\min \alpha$  recovers to 0.033. The longer  $L = 128$  run reaches comparable collapse depth and loses the proxy by  $t \simeq 78.4$ . We call this collapse-rebound morphology a *phantom bounce*; the term does not imply that a true apparent horizon has been located.

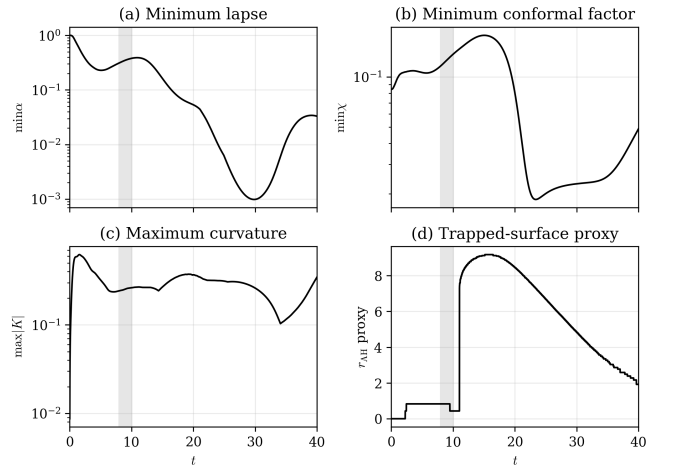


FIG. 2. Headline collapse diagnostics. The support ramp is shaded. The lapse and  $\chi$  pinch as curvature grows; the trapped-surface proxy then forms and subsequently shrinks.

## B. Gravitational-wave signal

The dominant  $r\Psi_4^{2,0}$  signal is coherent across extraction radii and its peak delay is consistent with propagation at  $c$  to the post-processing cadence. Its  $m = 0$  dominance means that the principal quadrupole follows the changing toroidal density; it is not evidence for an  $m = 2$  bar mode.

Figure 3(a) compares the three resolutions at  $R = 12$ . The medium and fine curves closely agree, but the coarse post-processing record has no samples over part of the principal burst. We therefore do not assign a three-point Richardson order to the burst amplitude. In the large box,  $R = 36$  and  $R = 48$  align in retarded time with overlap 0.993 after a residual shift of 0.35 and a relative  $L_2$  residual of 12.3% [Fig. 3(b)]. This supports approximate  $1/r$  behavior in the intermediate zone. Late growth at  $R = 36$  is not shared by  $R = 48$  and coincides with constraint growth, so it is not interpreted as a runaway gravitational-wave instability.

## C. Constraints and controls

For the headline run, the volume-weighted norms remain below  $\|\mathcal{H}\|_2 = 3.25 \times 10^{-2}$  and  $\|\mathcal{M}\|_2 = 9.67 \times 10^{-3}$  through  $t = 40$  (Fig. 4). The large-box norms are smaller than these values through  $t = 40$  but later reach 0.155 and 0.0417, respectively. Consequently its late waveform is treated cautiously.

No-ramp and delayed-ramp controls do not produce the same resolved deep-collapse branch before numerical failure. A nonrotating run built from the same toroidal modulus follows a pathological trajectory: its coordinate-sphere proxy expands, momentum constraints grow rapidly, and AMR memory approaches exhaustion. It was stopped at  $t \simeq 13.5$ . Thus the present data do not isolate rotation from toroidal geometry; the measured  $m = 0$  dominance only shows that rotation is not radiated mainly through nonaxisymmetric  $\ell = 2$  modes.

## D. Resolution study

For unequal spacings we assume, only where supported,

$$Q(h) = Q_0 + Ch^p \quad (7)$$

and solve

$$\frac{Q_c - Q_m}{Q_m - Q_f} = \frac{h_c^p - h_m^p}{h_m^p - h_f^p} \quad (8)$$

for  $p > 0$ . Sign-changing differences or unphysical continuum estimates are reported as non-asymptotic.

All three runs reproduce the collapse, peak proxy radius, shrinkage, lapse recovery, and axisymmetric burst

(Fig. 5). Selected scalar measures admit formal orders:

| $Q$                      | $p$  | $Q_0$                 | $\text{GCI}_f$ |
|--------------------------|------|-----------------------|----------------|
| $\max  K $               | 3.78 | 0.663                 | 3.31%          |
| $\max r_{\text{AH}}$     | 0.91 | 9.39                  | 2.39%          |
| $t(\max r_{\text{AH}})$  | 2.09 | 14.68                 | 2.42%          |
| $\max \ \mathcal{M}\ _2$ | 1.82 | $9.45 \times 10^{-3}$ | 1.95%          |

The Hamiltonian maximum instead increases slightly,  $0.0310 \rightarrow 0.0325 \rightarrow 0.0340$ , and has no positive observed order. The lapse and  $\chi$  minima deepen with refinement but extrapolate to unphysical negative values under the one-term model. These quantities establish qualitative multi-resolution robustness, not a universal continuum order.

## IV. DISCUSSION

Withdrawing support from this rotating exotic throat produces a trapped-surface proxy, a collapsing lapse, and a coherent  $\ell = 2$  wave without an imposed kick. The same collapse-rebound morphology appears at three resolutions and in a larger domain. This removes the strongest single-resolution and small-box concerns, but it does not turn every observable into a precision result.

Three limitations define the claim. First,  $r_{\text{AH}}$  is a coordinate-sphere proxy; a production apparent-horizon finder is needed for horizon mass and spin. Second, the waveform is extracted at finite, intermediate radii, and the coarse record misses part of the main burst. Third, active support is an external, nonconservative source, and whole-domain charge and angular momentum budgets do not close. The result is therefore a controlled numerical experiment on an engineered spacetime.

Future work should obtain a clean nonrotating control, add a genuine apparent-horizon finder, repeat the large box at higher AMR depth, and use wave-zone or characteristic extraction. These steps are required before quoting precision energies, quasinormal frequencies, or detector horizons.

## V. CODE AND DATA AVAILABILITY

The evolution example, initial-data solver, run parameters, validated `small_data/psi4_mode_l2m0.dat` files, diagnostics, and figure scripts are maintained in the project repository. The manuscript directory contains only the figures used here.

## VI. ACKNOWLEDGEMENTS

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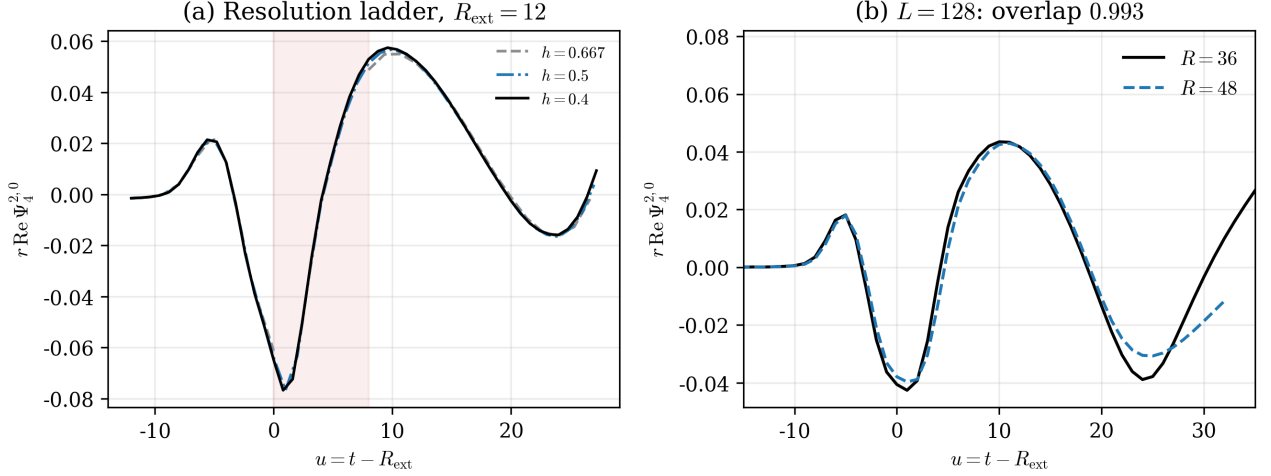


FIG. 3. **(a)** Validated  $r\Psi_4^{2,0}$  at  $R_{\text{ext}} = 12$  for the resolution ladder; the shaded region includes the coarse extraction gap and is excluded from formal waveform convergence. **(b)** Retarded-time comparison at  $R = 36, 48$  in the  $L = 128$  run. The agreement is an intermediate-zone cross-check, not a fourth resolution rung.

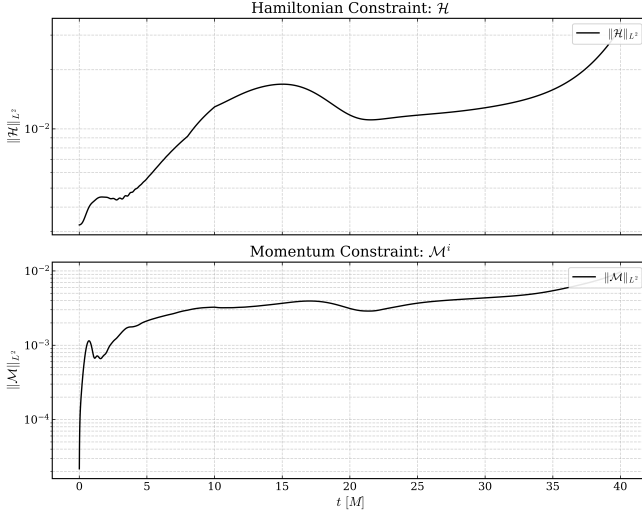


FIG. 4. Volume-weighted Hamiltonian and momentum constraint norms for the  $h = 0.5$  headline run. They remain bounded through the collapse and emission interval.

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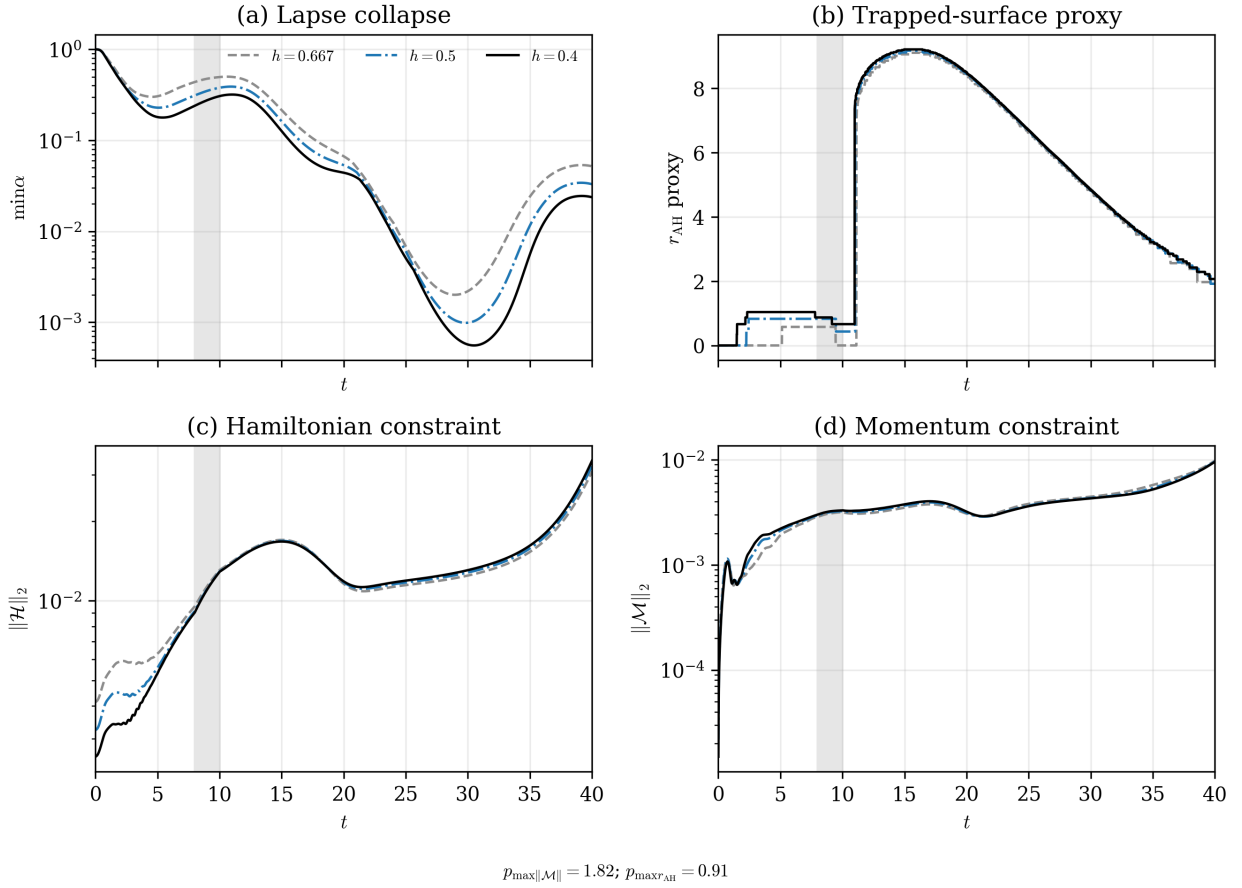


FIG. 5. Resolution ladder at  $h = 0.667, 0.5, 0.4$ . The support ramp is shaded. Collapse and proxy shrinkage are common to all runs. Momentum constraints show limited convergence; Hamiltonian maxima remain bounded but do not converge monotonically.